PYTHON LAB PROGRAMS

Flowcharts for all 10 tasks

Student Name: _____

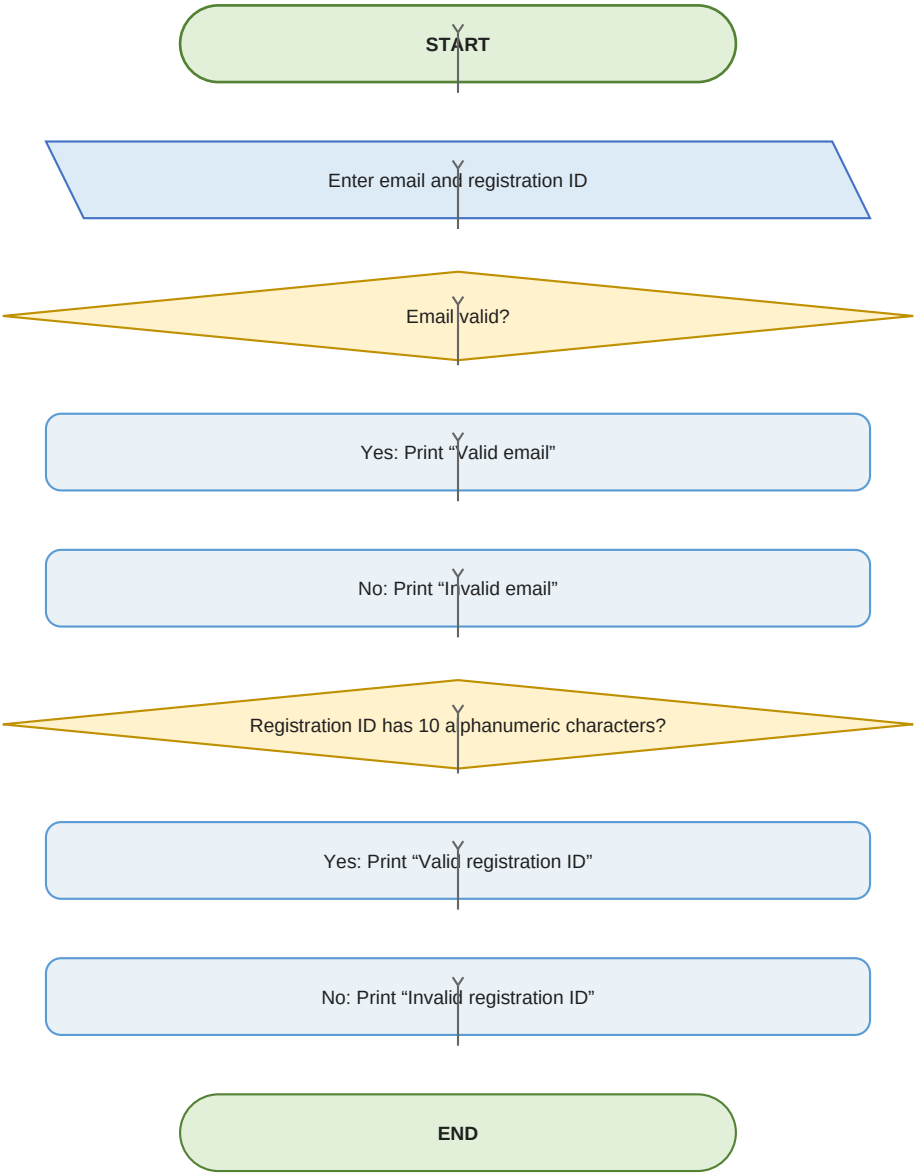
Registration Number: _____

Department / Branch: _____

College: _____

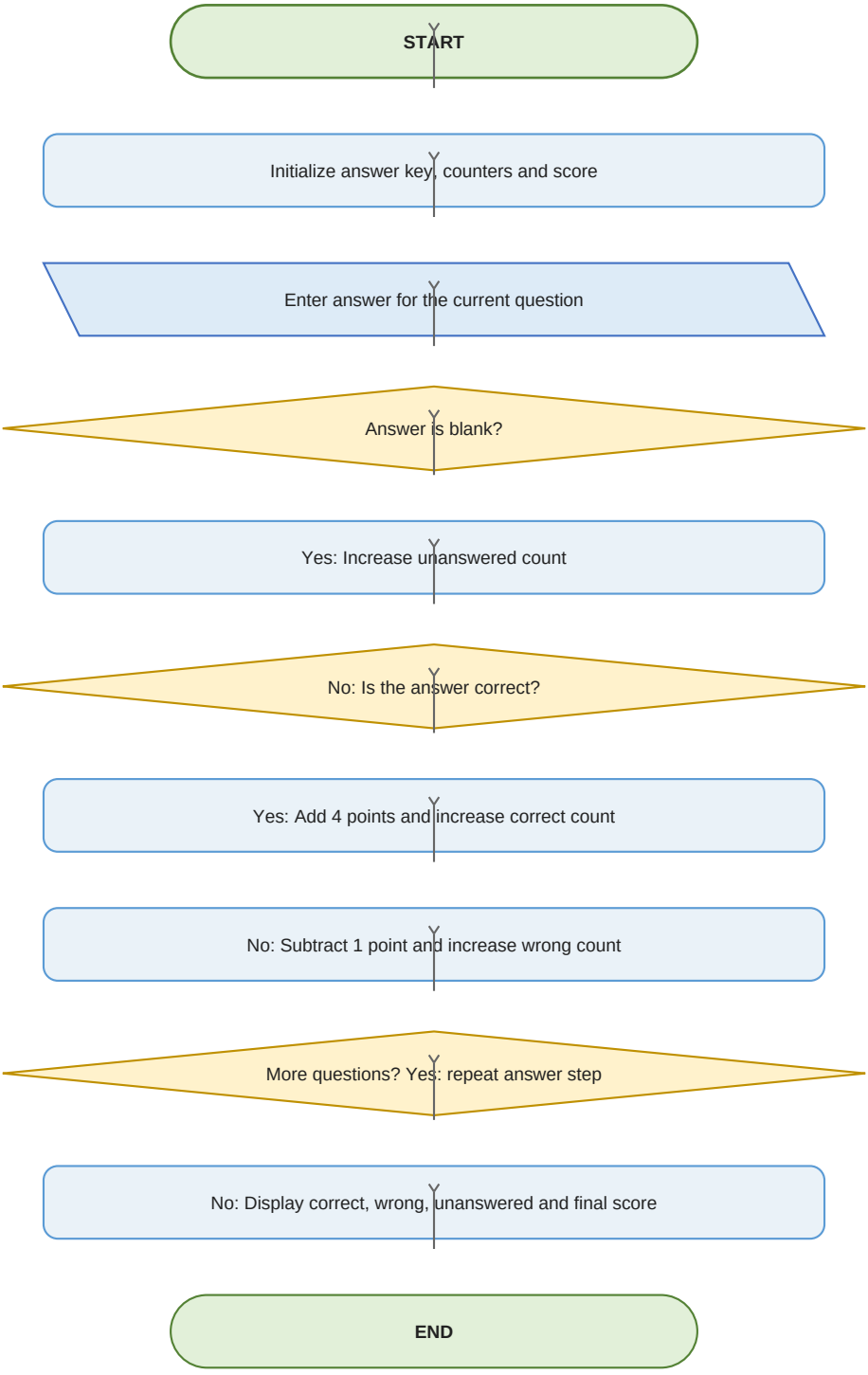
Flowchart key: Oval = Start/End; rectangle = process; parallelogram = input; diamond = decision.

Task 1 – Student Email and Registration ID Validation



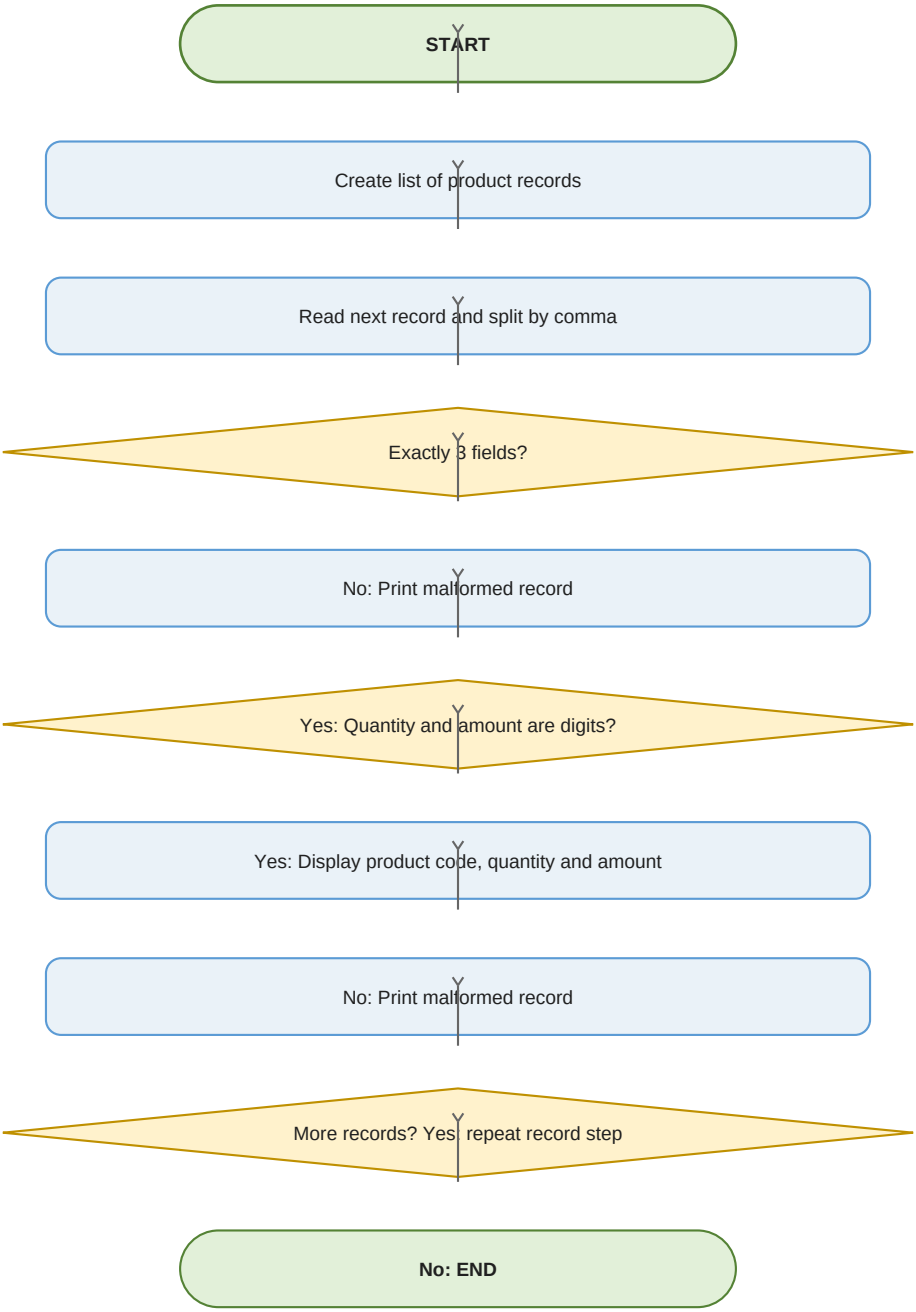
Note: Email and registration ID are checked independently. The email check is a basic format check.

Task 2 – Multiple-Choice Quiz Evaluation



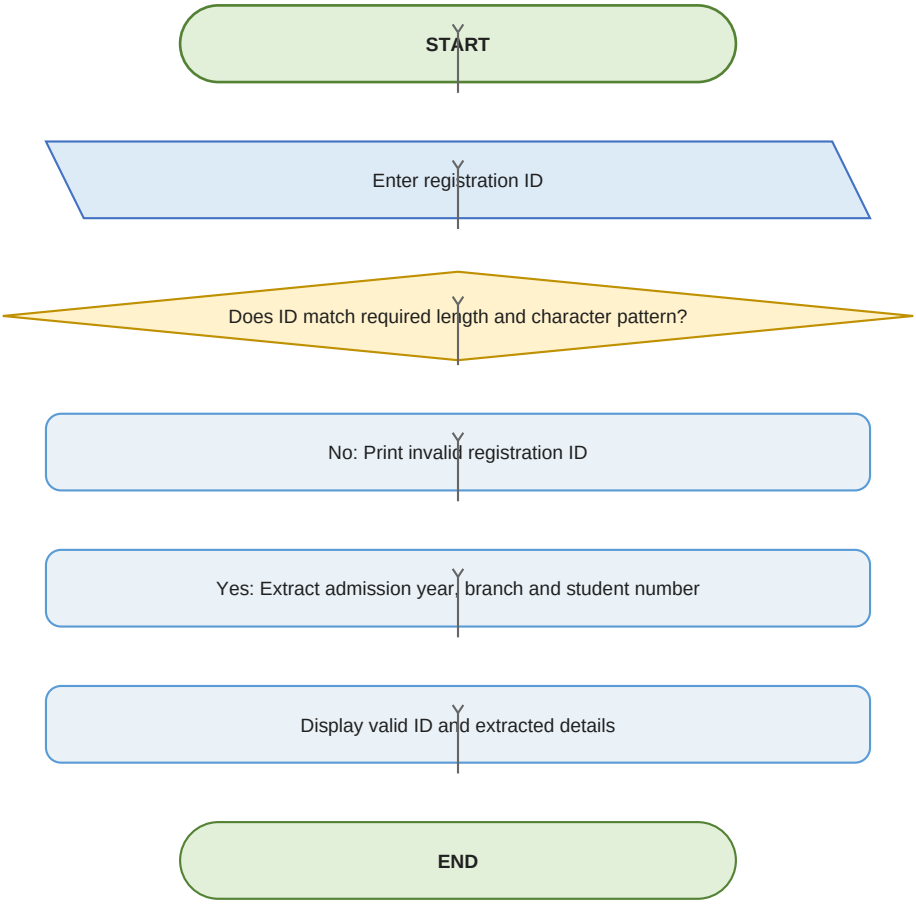
Note: Five questions are evaluated. Blank answers score 0, correct answers +4, and wrong answers –1.

Task 3 – Product Record Parsing and Validation



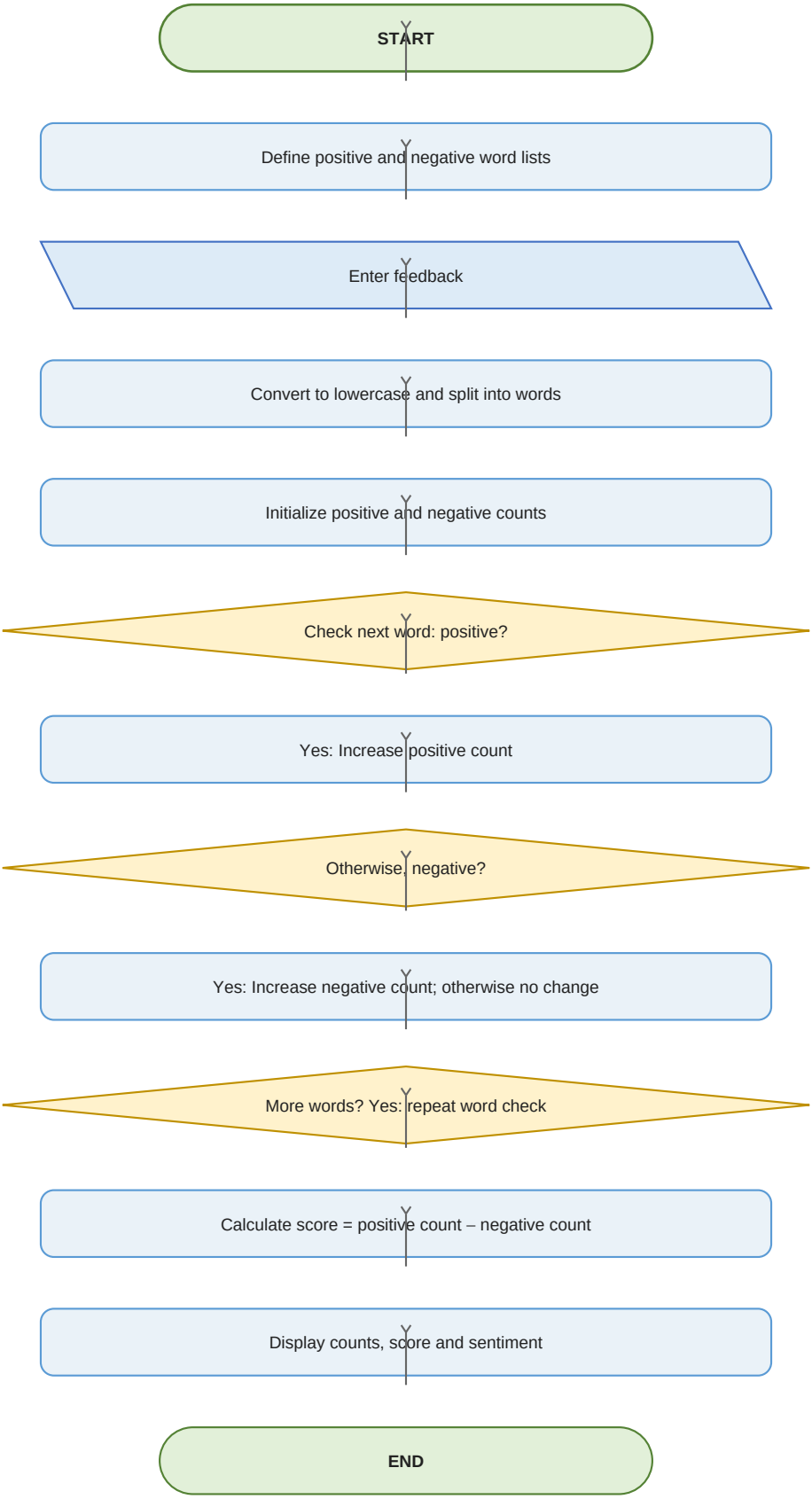
Note: Each comma-separated record must contain a product code, quantity and amount.

Task 4 – Registration ID Format Validation



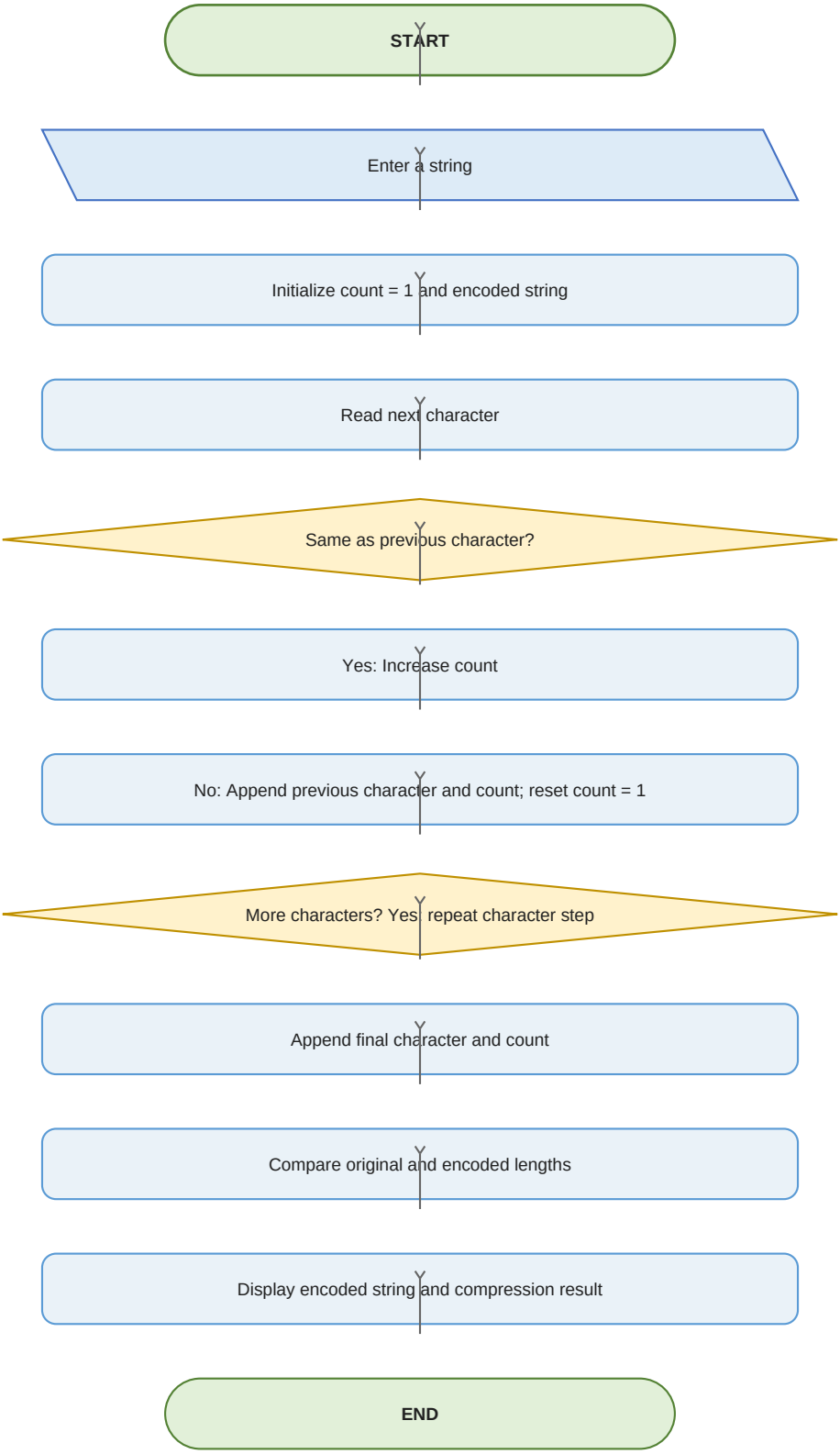
Note: The screenshot's example 24CSE101 has 8 characters, but the code checks for 9. Adjust the length and slices to match the intended format.

Task 5 – Feedback Sentiment Analysis



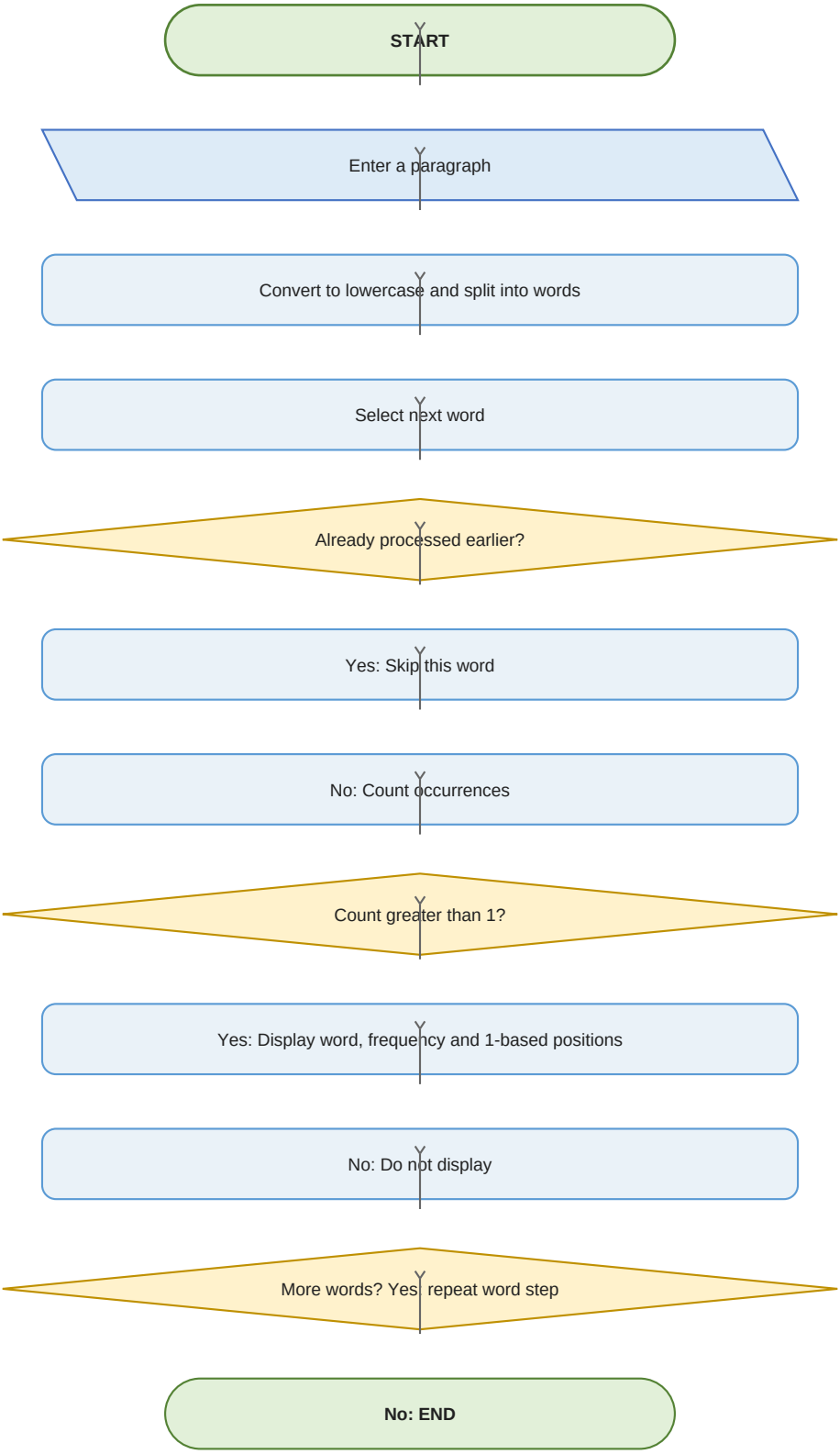
Note: The sentiment score is the number of positive words minus the number of negative words.

Task 6 – Run-Length Encoding



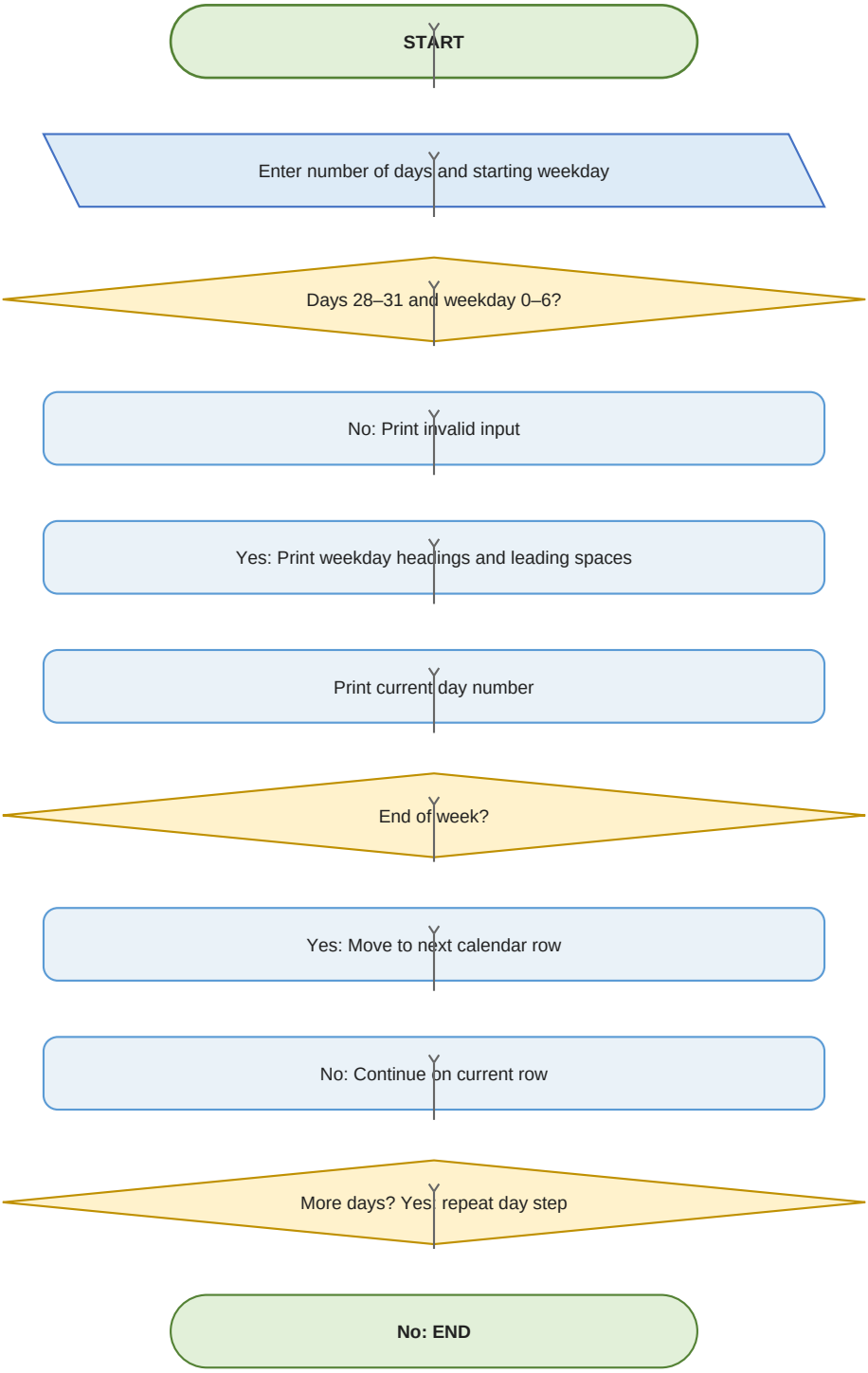
Note: Example: aaabb becomes a3b2. Encoding is beneficial only when the encoded string is shorter.

Task 7 – Word Frequency and Position Finder



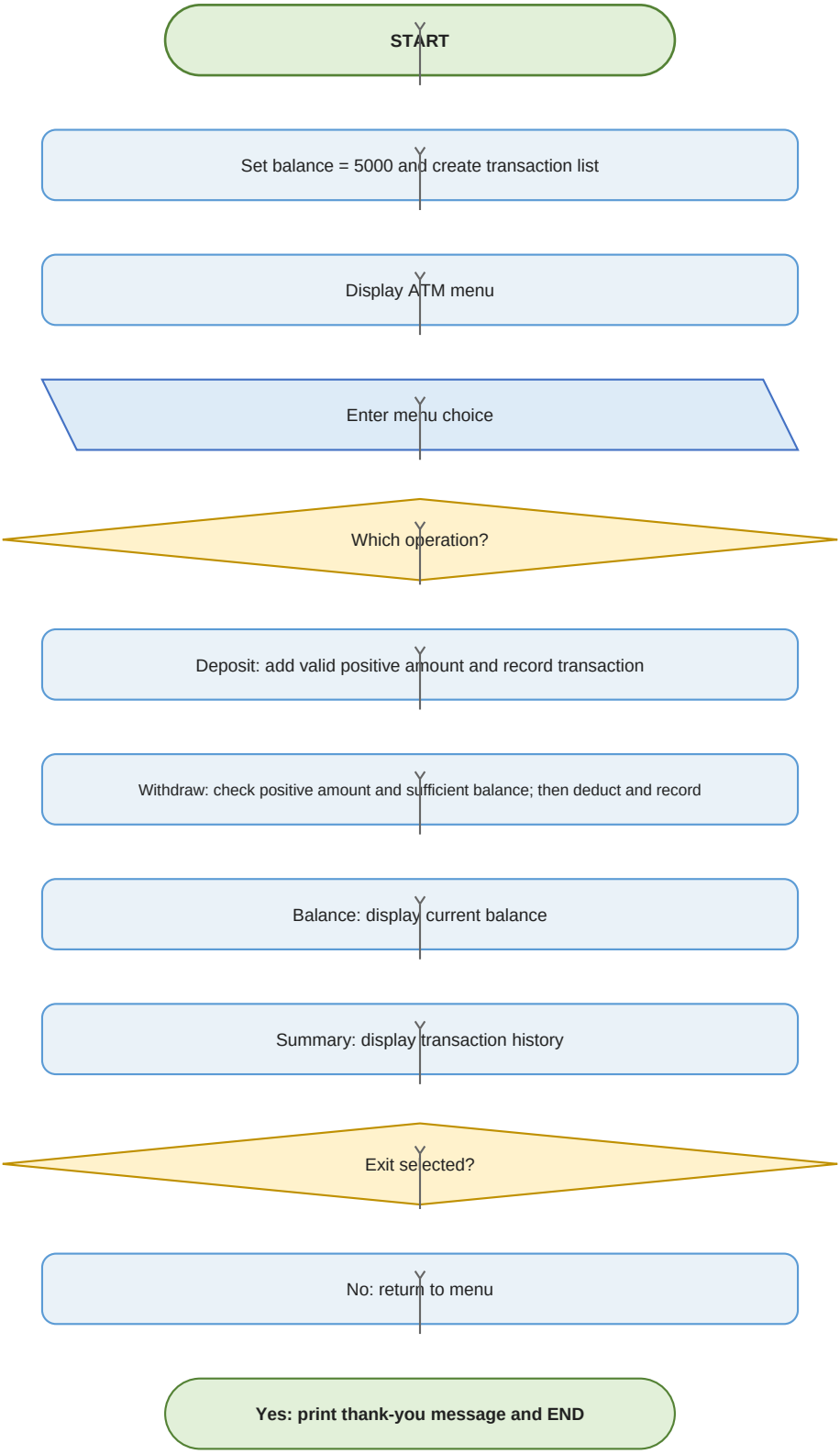
Note: Punctuation is not removed in the shown code, so words such as hello and hello, may be treated separately.

Task 8 – Calendar Layout Generator



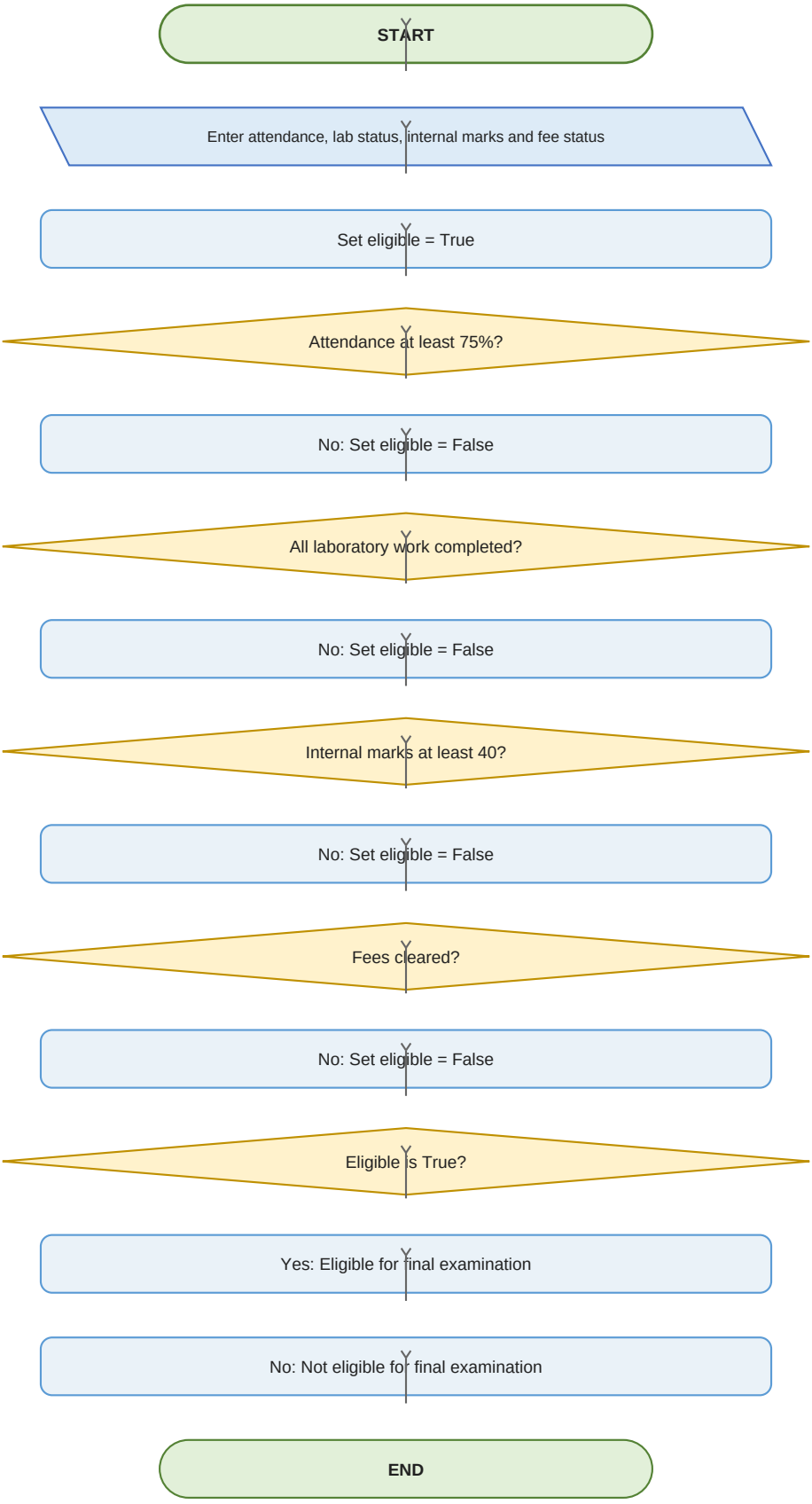
Note: Weekday numbering: 0 = Monday and 6 = Sunday.

Task 9 – Simple ATM Simulation



Note: Withdrawals must not exceed the available balance. Deposit and withdrawal amounts should be positive.

Task 10 – Final Examination Eligibility



Note: All four conditions must be satisfied: attendance ≥ 75%, labs completed, internal marks ≥ 40 and fees cleared.